25. Probability

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Probability as fraction/decimal/percentage:

A bag contains 5 red, 2 blue and 3 green counters. A counter is picked at random. Write the probability it is green as a fraction, decimal and percentage.

Answer: There are 10 counters in total, 3 of which are green, so P(green) = 3/10 = **0.3 = 30%**.

Example 2 — Probability of NOT happening:

The probability that it rains tomorrow is 0.4. What is the probability that it does NOT rain?

Answer: P(not rain) = 1 − 0.4 = **0.6**.

Example 3 — Combined events:

A fair coin is flipped twice. What is the probability of getting two tails?

Answer: List outcomes: HH, HT, TH, TT (4 equally likely outcomes). Only 1 is TT, so P(two tails) = **1/4**.

Example 4 — Without replacement:

A bag has 5 red and 5 blue counters. A red counter is picked and NOT put back. What is the probability the second counter picked is also red?

Answer: After removing 1 red, there are 4 red and 5 blue left, 9 counters in total. P(2nd is red) = **4/9**.

Example 5 — Multi-step: two dice:

Two fair dice are rolled and the scores are added together. What is the probability that the sum equals 10?

Answer: The ways to make 10 are (4,6), (5,5), (6,4) — 3 ways out of 36 equally likely outcomes. P(sum = 10) = 3/36 = **1/12**.

Foundation

15 questions

1. A fair six-sided dice is rolled once. What is the probability of rolling a 5? Give your answer as a fraction.

2. A spinner has 6 equal sections numbered 1 to 6. What is the probability of landing on an even number?

3. Write the probability 'impossible' as a number. A) 0 B) 0.25 C) 0.5 D) 0.75 E) 1

4. A bag contains 10 balls, 7 of which are red. What is the probability of picking a red ball, as a decimal?

5. Order these words from least likely to most likely: likely, impossible, evens, certain, unlikely.

6. A coin is flipped once. What is the probability of landing on heads, as a percentage?

7. Which word best describes an event with probability 0? A) Impossible B) Unlikely C) Evens D) Likely E) Certain

8. A bag contains only red counters. A counter is picked at random. What is the probability it is red?

9. List all the possible outcomes when a single fair six-sided dice is rolled.

10. A box contains 4 blue pens and 6 green pens. What is the probability of picking a blue pen, as a fraction in its simplest form?

11. Select ALL that apply: which of these are valid probabilities for an event? A) 0 B) 1.5 C) 0.3 D) −0.1 E) 1

12. A fair six-sided dice is rolled once. What is the probability of rolling a number greater than 3? A) 1/6 B) 1/3 C) 1/2 D) 2/3 E) 5/6

13. The probability that a train is late is 7/20. Write this probability as a percentage.

14. List all the possible outcomes when a spinner numbered 1 to 4 is spun.

15. A spinner is divided into 5 equal sections coloured red, blue, green, yellow and purple. What is the probability of NOT landing on blue?

Standard

20 questions

1. A bag contains 6 green, 2 red and 2 yellow counters. A counter is picked at random. What is the probability it is NOT green? Give your answer as a fraction.

2. A spinner has 10 equal sections: 3 are red, 7 are white. What is the probability of landing on red, as a decimal?

3. The probability that Zara wins a game is 0.6. What is the probability that she loses (assuming no draws)? A) 0.2 B) 0.3 C) 0.4 D) 0.5 E) 0.6

4. A fair coin is flipped twice. List all 4 possible outcomes, then state the probability of getting exactly one head and one tail.

5. A bag contains 25 sweets: 10 are orange, the rest are lemon. What is the probability of picking a lemon sweet, as a percentage?

6. Two fair coins are flipped. What is the probability of getting at least one tail? A) 1/4 B) 1/2 C) 3/4 D) 1 E) 0

7. A bag has 5 red and 5 blue counters. A counter is picked, its colour noted, and it is NOT put back. If the first counter picked is red, what is the probability the second counter picked is also red?

8. Select ALL that apply: a fair six-sided dice is rolled once. Which statements are true? A) P(rolling a 7) = 0. B) P(rolling an even number) = 1/2. C) P(rolling a number less than 7) = 1. D) P(rolling a 1) = 1/6. E) P(rolling a multiple of 5) = 1/3.

9. A box of 20 sweets has 12 milk and 8 dark chocolates. What is the probability of picking a milk chocolate, as a fraction in its simplest form?

10. A weather forecaster says there is a 60% chance of rain tomorrow. What is the probability, as a fraction, that it will NOT rain?

11. A bag contains only red and blue counters. P(red) = 0.4. If there are 30 counters in total, how many are blue?

12. A fair spinner has sections numbered 1 to 4. It is spun twice. What is the probability that it lands on 4 both times? A) 1/16 B) 1/8 C) 1/4 D) 1/2 E) 1/2

13. A drawer contains 5 black socks and 5 white socks. One sock is pulled out at random, not put back, and a second sock is pulled out. What is the probability that both socks are black?

14. In a raffle, 150 tickets are sold and Omar buys 3 of them. What is the probability that Omar's ticket is drawn as the winner, as a decimal?

15. Select ALL that apply: a bag contains 4 red, 3 blue and 3 green counters (10 total). Which statements are true? A) P(red) = 2/5. B) P(not blue) = 7/10. C) P(green) = P(blue). D) P(blue) = 0.4. E) P(red or green) = 0.7.

16. A bag contains 7 blue and 3 red counters. What is the probability of picking a red counter, as a decimal?

17. A spinner numbered 1 to 6 is spun once. What is the probability of landing on a multiple of 3?

18. A bag has 8 red and 4 blue counters. A counter is picked, its colour noted, and it is NOT put back. If the first counter picked is red, what is the probability the second counter picked is also red?

19. A spinner numbered 1 to 8 is spun once. What is the probability of landing on a multiple of 3 or a multiple of 4, as a fraction?

20. A box of 24 sweets has 9 toffees, and the rest are mints. What is the probability of picking a mint, as a fraction in its simplest form?

Hard

5 questions

1. In a jar of red, blue and green buttons, exactly half are red and a third are blue. Given that the jar holds 24 buttons altogether, how many buttons are green?

2. Roll two fair dice and add the two scores together. What is the probability that this sum equals 9? Write your answer as a fraction.

3. A box contains 5 red counters and 3 blue counters. One counter is drawn out and kept to one side, then a second is drawn. Find the probability that both counters drawn are blue.

4. Spin a 1-to-5 spinner once and toss a fair coin once. What is the probability that the spinner lands on an even number and the coin lands on tails, both happening together?

5. Select ALL that apply: a box contains 12 cards numbered 1 to 12, and one card is drawn at random. Which statements are true? A) P(drawing an even number) = 1/2. B) P(drawing a multiple of 4) = 1/4. C) P(drawing a number greater than 10) = 1/6. D) P(drawing a square number) = 1/3. E) P(drawing a prime number) = 5/12.

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